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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Event Registration and Participant Management System

Submitted by

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APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled "**PROJECT TITLE"** submitted in partial fulfillment of the requirements for the course [B.c COMPUTER SCIENCE**]** is a **record of my original work** carried out under the guidance of [ARVIND], Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

NITHRESH R

(Your Name)

CERTIFICATE

This is to certify that the project entitled" **Event Registration and Participant Management System"** is a **bona fide work** carried out by NITHRESH R, B.Sc. COMPUTER SCIENCE ,**KG College of Arts & Science**, in partial fulfillment of the requirements for the course java programming.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:10.04.2026

Submitted for the Viva-Voce Examination held on 10.04.2024

Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

ARVIND BOOPALAN

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ABSTRACT

The Event Registration and Participant Management System is a software application developed to automate the process of event registration and participant data management. In traditional systems, event registration is carried out manually using paper forms or spreadsheets, which often results in errors, duplication of data, and inefficient record maintenance. These problems become more serious when the number of participants increases.

This project aims to provide a computerized solution using Java programming language to handle event registration in an efficient and organized manner. The system allows users to register for events by entering their personal details such as name, contact information, and event preferences. The entered data is stored and can be accessed, updated, or deleted by the administrator.

The system is developed using Core Java concepts including object-oriented programming (OOP), classes, objects, methods, loops, arrays, and conditional statements. The application can be implemented as a console-based program or extended into a graphical user interface using Java Swing.

The main goal of this project is to reduce manual work, improve data accuracy, and provide a structured approach to event management. This system also serves as a basic model for developing advanced event management systems with database integration and online registration capabilities.

1. INTRODUCTION

In today's fast-paced digital world, event management has become an essential activity in educational institutions, corporate organizations, and social communities. Events such as seminars, workshops, conferences, and cultural programs require proper planning and efficient participant management.

Traditionally, event registration is handled manually through paper forms or simple spreadsheets. This approach is not only time-consuming but also prone to human errors such as incorrect data entry, duplication of records, and loss of information. Managing large numbers of participants becomes very difficult using manual methods.

To overcome these issues, computerized systems are widely used. An Event Registration and Participant Management System provides a digital platform where users can register for events easily, and administrators can manage participant data effectively. The system ensures that all records are stored systematically and can be accessed whenever required.

This project is designed using Java programming language and demonstrates how software can simplify real-world problems. It focuses on improving efficiency, accuracy, and reliability in event registration processes.

1.1 Objective

The primary objective of this project is to develop a simple and efficient system for managing event registrations using Java.

The detailed objectives are:

- To create a digital system for event registration
- To store participant details in an organized manner

- To reduce errors caused by manual data entry
- To provide easy access to participant information
- To allow updating and deletion of records
- To demonstrate the use of object-oriented programming concepts
- To improve efficiency and reduce time consumption

1.2 Overview

The Event Registration and Participant Management System is a menu-driven application developed using Java. The system provides an interface where users can perform various operations related to event registration.

Users can enter their personal details and select the event they wish to participate in. The system stores this information and allows the administrator to manage the data effectively.

The application includes multiple functionalities such as adding new participants, viewing participant details, updating existing records, and deleting records if required. The system is designed in such a way that it can handle multiple participants efficiently.

This project demonstrates how programming can be applied to real-life scenarios and provides a foundation for developing more advanced systems with database connectivity and web-based interfaces.

1.3 Features

The system includes the following features:

- **User Registration**
  Allows participants to enter their personal details such as name, ID, and contact information.
- **Event Selection**
  Participants can choose from available events.
- **Data Storage**
  Stores participant details using arrays or data structures.
- **View Participants**
  Displays all registered participants.
- **Search Functionality**
  Allows searching participant details using ID or name.
- **Update Records**
  Enables modification of participant information.
- **Delete Records**
  Removes participant details from the system.
- **User-Friendly Interface**
  Simple and easy-to-use menu-driven system.
- **Efficiency**
  Reduces time and effort compared to manual systems.

1. PROGRAM ANALYSIS

Program analysis is a crucial step in software development where the problem is studied in detail and requirements are identified.

2.1 Problem Definition

Manual event registration systems have several drawbacks:

- High chances of data entry errors
- Duplicate registrations
- Difficulty in managing large datasets
- Time-consuming processes
- Lack of proper record maintenance
- Risk of data loss

These problems highlight the need for an automated system that can efficiently manage event registrations and participant data.

2.2 Project Overview

The system is designed to provide a complete solution for managing event registrations. It includes functionalities such as:

- Registering new participants
- Storing participant details
- Displaying participant information
- Updating and deleting records

The system ensures accuracy and efficiency by automating all operations.

2.3 Scopes and Limitations

Scope

- Managing event registrations
- Handling participant data
- Providing organized records

Applications:

- Colleges and universities
- Corporate events
- Workshops and seminars
- Cultural programs

Limitations:

- Limited to basic functionality
- No online access (in simple version)
- No database integration (if not implemented)

1. SYSTEM DESIGN

System design defines how the system is structured and how different components interact

3.1 Modules

The main modules of the **Hotel & Restaurant Billing System** are explained below:

1. Registration Module

This module collects participant details such as name, ID, and event selection. It ensures that all required information is entered correctly.

2. Participant Management Module

This module displays participant details and helps in managing records efficiently.

3. Update/Delete Module

Allows modification and removal of participant data. This ensures data accuracy and flexibility.

4. Event Selection Module

Provides a list of events and allows users to select their preferred event.

5. Exit Module

Safely terminates the application.

1. METHODOLOGY

Methodology refers to the systematic process followed to design, develop, and implement the Event Registration and Participant Management System. It explains the step-by-step approach used in transforming the problem into a working software solution.

A structured software development approach is followed to ensure that the system is efficient, reliable, and meets all user requirements. The methodology used in this project includes several important phases such as problem analysis, system design, program development, testing, and documentation.

Each phase plays a crucial role in the successful completion of the project.

4.1. Problem Analysis

Problem analysis is the first and most important step in developing the system. In this phase, the existing manual system is carefully studied to identify its drawbacks and limitations.

In traditional event registration systems, data is recorded manually using paper forms or spreadsheets. This leads to several issues such as:

- Errors in data entry due to human mistakes
- Duplicate registrations of participants
- Difficulty in storing and retrieving large amounts of data
- Time-consuming registration process
- Lack of proper organization of records
- Risk of data loss or misplacement

These problems create inefficiency and increase the workload for event organizers.

After analyzing these issues, the need for a computerized system is identified. The proposed system should be able to automate the registration process, store participant data securely, and provide easy access to information.

Thus, the requirements of the system are clearly defined in this phase.

4.2. System Design

System design is the process of planning the structure and components of the system before implementation. It acts as a blueprint for developing the application.

In this phase, the system is divided into different modules, each responsible for a specific function. The main aim is to design a system that is simple, efficient, and user-friendly.

The following design decisions are made:

- The system will be menu-driven for easy interaction
- Data will be stored using arrays or objects
- Separate modules will handle registration, viewing, updating, and deletion
- Input and output operations will be clearly defined

The design also includes planning the flow of the program, which ensures smooth execution of tasks. Proper system design reduces complexity and helps in easy implementation and maintenance of the program.

4.3. Program Development

Program development is the phase where the actual coding of the system takes place. Based on the system design, the program is implemented using the Java programming language.

The development process involves:

- Creating classes to represent participants and events
- Defining variables to store participant details
- Writing methods for different operations such as registration, display, update, and deletion
- Implementing loops to allow repeated operations
- Using conditional statements to control program flow

Core Java concepts such as object-oriented programming (OOP), encapsulation, and modular programming are used to build the system.

The program is designed in a structured manner so that each module performs its specific task efficiently. Proper coding practices are followed to ensure readability and maintainability of the program.

4.4. Testing and Validation

Testing is an essential phase in software development that ensures the system works correctly and meets user requirements.

In this phase, the developed program is tested using different inputs to verify its functionality. The goal is to identify and fix errors or bugs in the system.

Different types of testing are performed:

- **Unit Testing** – Testing individual modules separately
- **Integration Testing** – Testing combined modules

- **System Testing** – Testing the entire system

Test cases are used to check:

- Correct registration of participants
- Accurate display of data
- Proper updating and deletion of records
- Handling of invalid inputs

Validation ensures that the system produces correct and expected results. This phase improves the reliability and performance of the application.

4.5. Result and Documentation

After successful testing, the results of the system are analyzed and documented.

The system is able to:

- Register participants successfully
- Store and manage participant data
- Display accurate information
- Perform update and delete operations

Documentation is prepared to explain the working of the system. It includes details such as system design, methodology, algorithm, and outputs.

Proper documentation helps users understand the system and also serves as a reference for future improvements.

4.6. Algorithm

An algorithm is a step-by-step procedure used to solve a problem. It provides a clear logic for the program.

Algorithm for Event Registration System:

1. Start the program
2. Display menu options
3. Accept user choice
4. If user selects registration:
    - Enter participant details
    - Store the data
5. If user selects view:
    - Display participant records
6. If user selects update:
    - Modify existing data
7. If user selects delete:
    - Remove participant record
8. Repeat the process until user exits
9. End the program

This algorithm ensures smooth execution of the system.

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start → Display Menu → Enter Choice → Perform Operation (Register/View/Update/Delete) → Display Result → End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

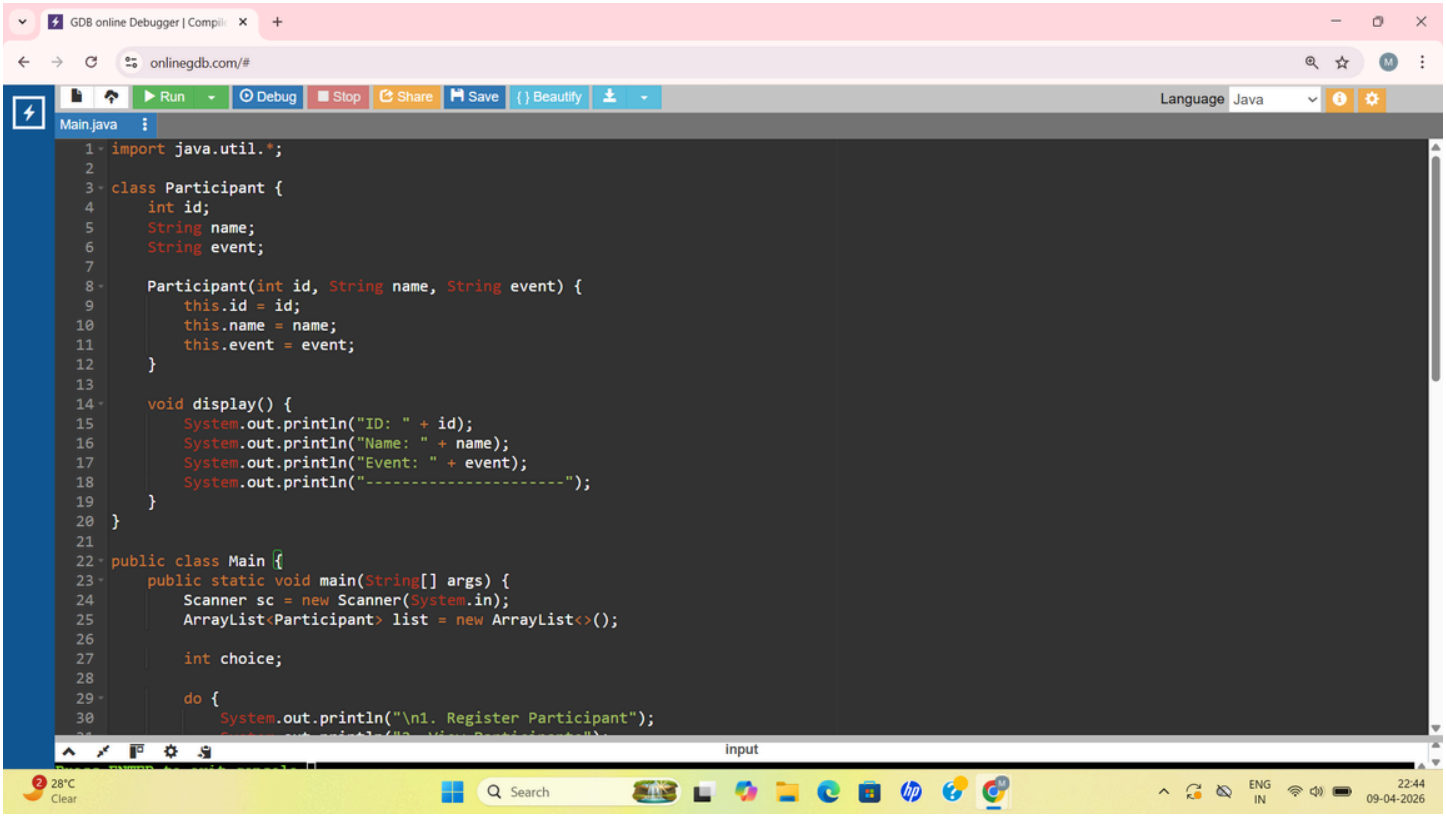
1. SOURCE CODE

A class named **Participant** is created to store participant details such as ID, name, and event. The main class contains the menu-driven program which allows users to register and view participants.

The following inputs are provided to the system:

- **User Choice:**
  The user selects an option from the menu such as Register Participant, View Participants, or Exit.
- **Participant ID:**
  A unique identification number is entered for each participant.
- **Participant Name:**
  The name of the participant is entered as text.
- **Event Name:**
  The user enters the name of the event in which the participant wants to register.

1. **INPUT AND OUTPUT SCREENS**




```java
import java.util.*;

class Participant {
    int id;
    String name;
    String event;

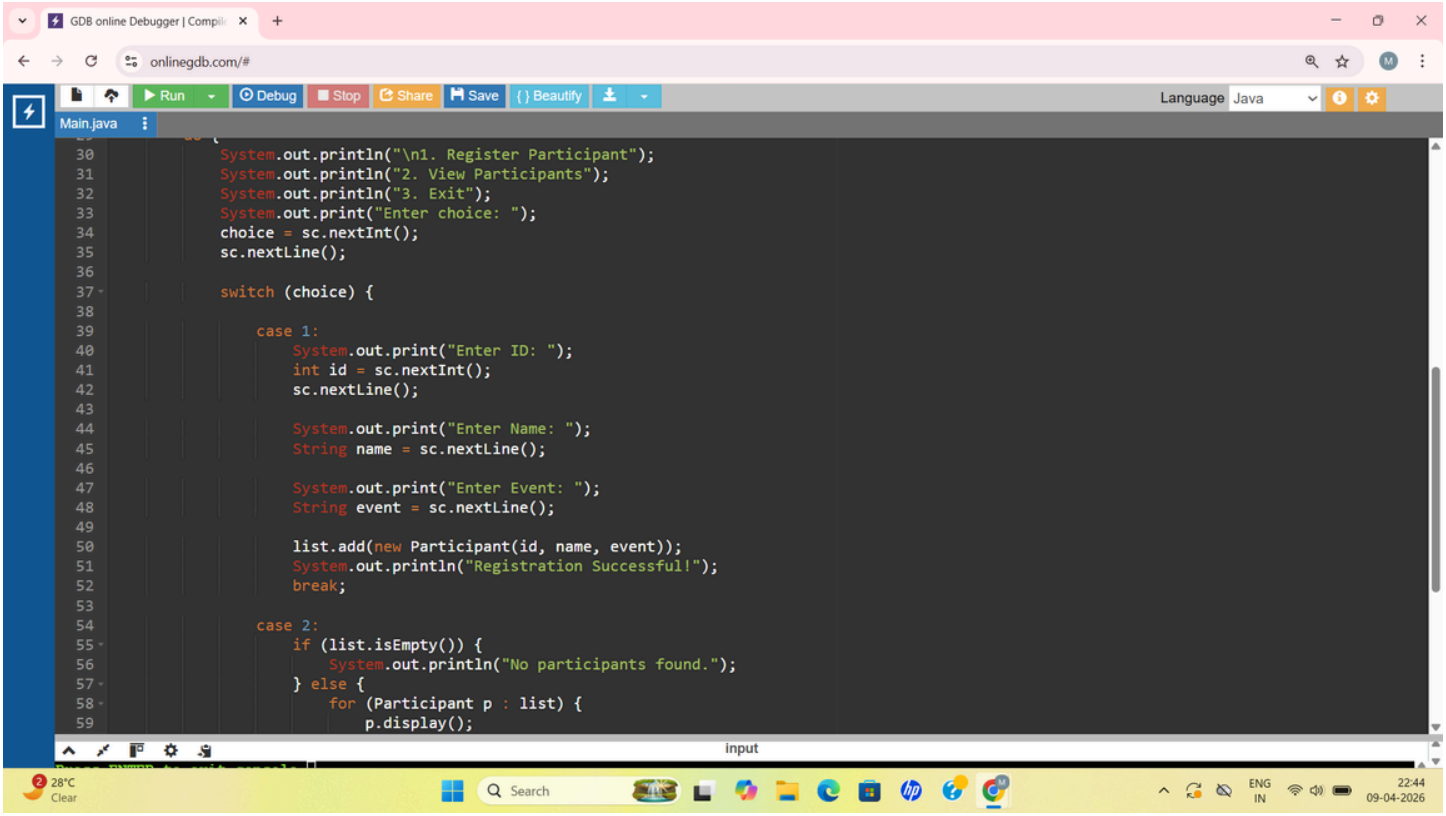
    Participant(int id, String name, String event) {
        this.id = id;
        this.name = name;
        this.event = event;
    }

    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Event: " + event);
        System.out.println("----------------------");
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Participant> list = new ArrayList<>();

        int choice;

        do {
            System.out.println("\n1. Register Participant");
```




```java
            System.out.println("\n1. Register Participant");
            System.out.println("2. View Participants");
            System.out.println("3. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {

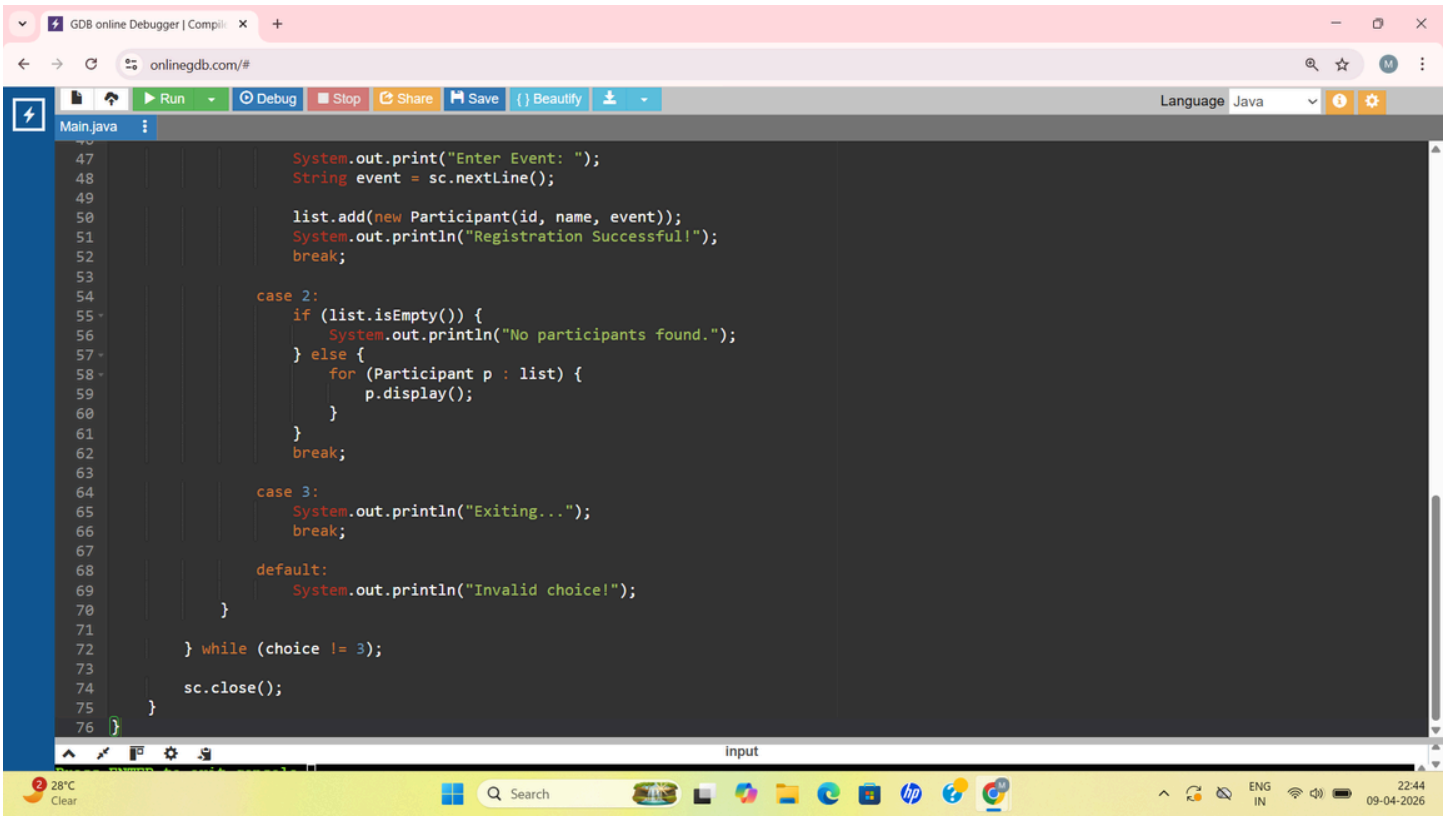
                case 1:
                    System.out.print("Enter ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();

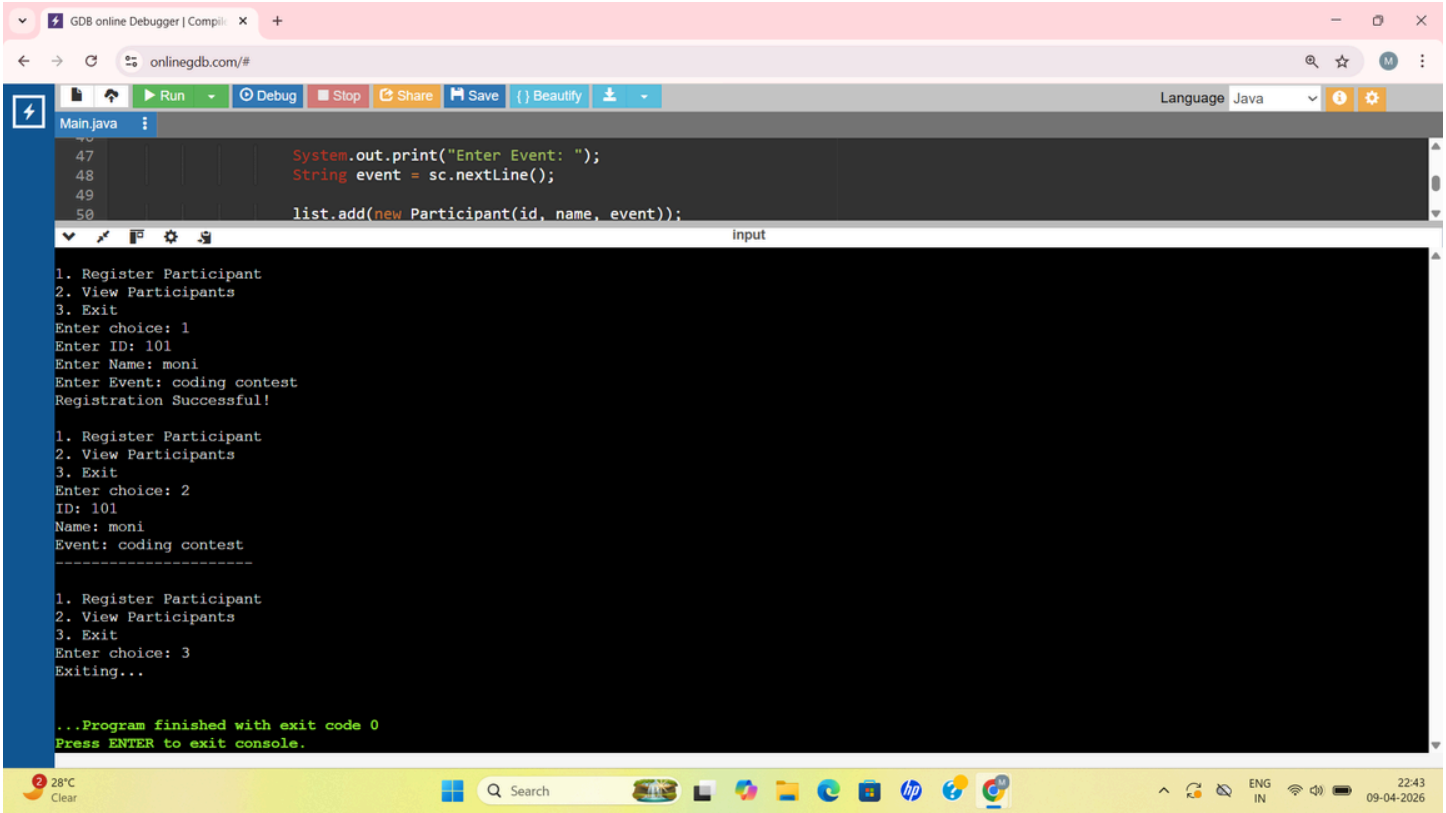
                    System.out.print("Enter Name: ");
                    String name = sc.nextLine();

                    System.out.print("Enter Event: ");
                    String event = sc.nextLine();

                    list.add(new Participant(id, name, event));
                    System.out.println("Registration Successful!");
                    break;

                case 2:
                    if (list.isEmpty()) {
                        System.out.println("No participants found.");
                    } else {
                        for (Participant p : list) {
                            p.display();
```


GDB online Debugger | Compil
onlinegdb.com/#
Run
Debug
Stop
Share
Save
{ } Beautify
Language
Java
Main.java
System.out.print("Enter Event: ");
String event = sc.nextLine();
list.add(new Participant(id, name, event));
System.out.println("Registration Successful!");
break;
case 2:
if (list.isEmpty()) {
System.out.println("No participants found.");
} else {
for (Participant p : list) {
p.display();
}
}
break;
case 3:
System.out.println("Exiting...");
break;
default:
System.out.println("Invalid choice!");
} while (choice != 3);
sc.close();
}
}
input
28°C
Clear
Search
ENG
IN
22:44
09-04-2026


GDB online Debugger | Compil
onlinegdb.com/#
Run
Debug
Stop
Share
Save
{ } Beautify
Language
Java
Main.java
System.out.print("Enter Event: ");
String event = sc.nextLine();
list.add(new Participant(id, name, event));
input
1. Register Participant
2. View Participants
3. Exit
Enter choice: 1
Enter ID: 101
Enter Name: moni
Enter Event: coding contest
Registration Successful!
1. Register Participant
2. View Participants
3. Exit
Enter choice: 2
ID: 101
Name: moni
Event: coding contest
1. Register Participant
2. View Participants
3. Exit
Enter choice: 3
Exiting...
...Program finished with exit code 0
Press ENTER to exit console.
28°C
Clear
Search
ENG
IN
22:43
09-04-2026

1. DISCUSSION & CONCLUSION

7.1 Discussion

The Event Registration and Participant Management System was developed to simplify the process of managing event registrations. The system allows users to register participants and manage their details efficiently.

It reduces manual work and minimizes errors such as duplicate entries and incorrect data handling. The menu-driven approach makes the system easy to use and understand.

However, the system has some limitations such as lack of database support and advanced features. These can be improved in future enhancements.

7.2 Conclusion

The Event Registration and Participant Management System is a simple and effective application developed using Java concepts. It helps in managing participant data in an organized way and reduces manual effort.

The project successfully meets its objectives and demonstrates the practical use of programming. It can be further improved by adding features like database integration and graphical interface.

1. REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*
- E. Balagurusamy – *Programming with Java*

Websites

https://www.oracle.com/java

https://www.geeksforgeeks.org

https://www.w3schools.com

https://www.javatpoint.com